4.2 The processes and features associated with earthquakes and volcanoes

This chapter will explain:

- ★ the processes at plate boundaries that cause earthquakes and volcanoes
- ★ the main characteristics of earthquakes
- ★ the main types of volcanoes
- ★ how volcanoes are classified
- ★ the main features of volcanoes
- ★ the hazards associated with volcanoes

The processes at plate boundaries that cause earthquakes and volcanoes

At divergent/constructive plate boundaries two plates move apart from each other, causing **sea floor spreading**. New oceanic crust is formed, creating mid-ocean ridges and pushing the plates apart; volcanic activity is common. Less viscous, runny basaltic lava forms shield mountains. A narrow belt of shallow focus, low-magnitude earthquakes is formed. An example is the Mid-Atlantic Ridge (Europe is moving away from Northern America).

At convergent/destructive plate boundaries the oceanic crust moves towards the continental crust and is subducted (sinks) beneath it due to its greater density, pulling the rest of the plate behind it. Deep-sea trenches and island arcs are formed. Volcanic activity is common. More acidic, explosive lava results in cone volcanoes. **Pyroclastic flows** and **lahars** may occur (Figure 4.9). The plate movement compresses and folds the seafloor rocks/sediments, forming fold mountains. Earthquakes are a mix of shallow-, intermediate- and deep-focus and can be high magnitude. An example is the Nazca Plate sinking under the South American plate.

At a convergent/collision boundary two continental crusts collide. As neither can be subducted (sink), the crust and sediments become crumpled up into fold mountains. There is no volcanic activity due to the lack of magma. Earthquakes are shallow focus but powerful due to the friction as the plates move. A good example is the Indian Plate colliding with the Eurasian Plate to form the Himalayas. Prior to this, the Tethys Sea (continental crust) had collided into the Eurasian Plate. However, once it had fully subducted, the Indian Plate that it was dragging behind it collided with the Eurasian Plate to form a collision boundary.

At a conservative/transform plate boundary, two plates slip sideways past each other, but land is neither destroyed nor created. There is no volcanic activity due to the lack of magma. Earthquakes are shallow focus. Stress can build up close to the surface as plates ‘get stuck’ and this is released as earthquakes. The San Andreas Fault in California is a good example of a conservative/transform plate boundary, where the Pacific Plate is slipping past the North American Plate.

▲ **Figure 4.9** The impact of lahars at Plymouth, Montserrat

Activities

1 State the types of plate boundaries where:

a volcanic activity occurs
b earthquakes occur.

2 Identify the type of location where volcanoes occur that is not a plate boundary.

The main characteristics of earthquakes

Earthquakes involve sudden, violent shaking of the Earth's surface. They occur after a build-up of pressure causes rocks and other materials to give way. Most of this pressure occurs at plate boundaries, when one plate is moving against another. Earthquakes are associated with all types of plate boundary.

The **focus** refers to the place beneath the ground where the earthquake takes place. **Deep-focus earthquakes** are associated with **subduction zones**. **Shallow-focus earthquakes** are generally located along divergent/constructive boundaries and along conservative/transform boundaries (Figure 4.10). The **epicentre** is the point on the ground surface immediately above the focus.

Some earthquakes are caused by human activity, including nuclear testing, building large dams, drilling for oil/natural gas (fracking) and coal mining.

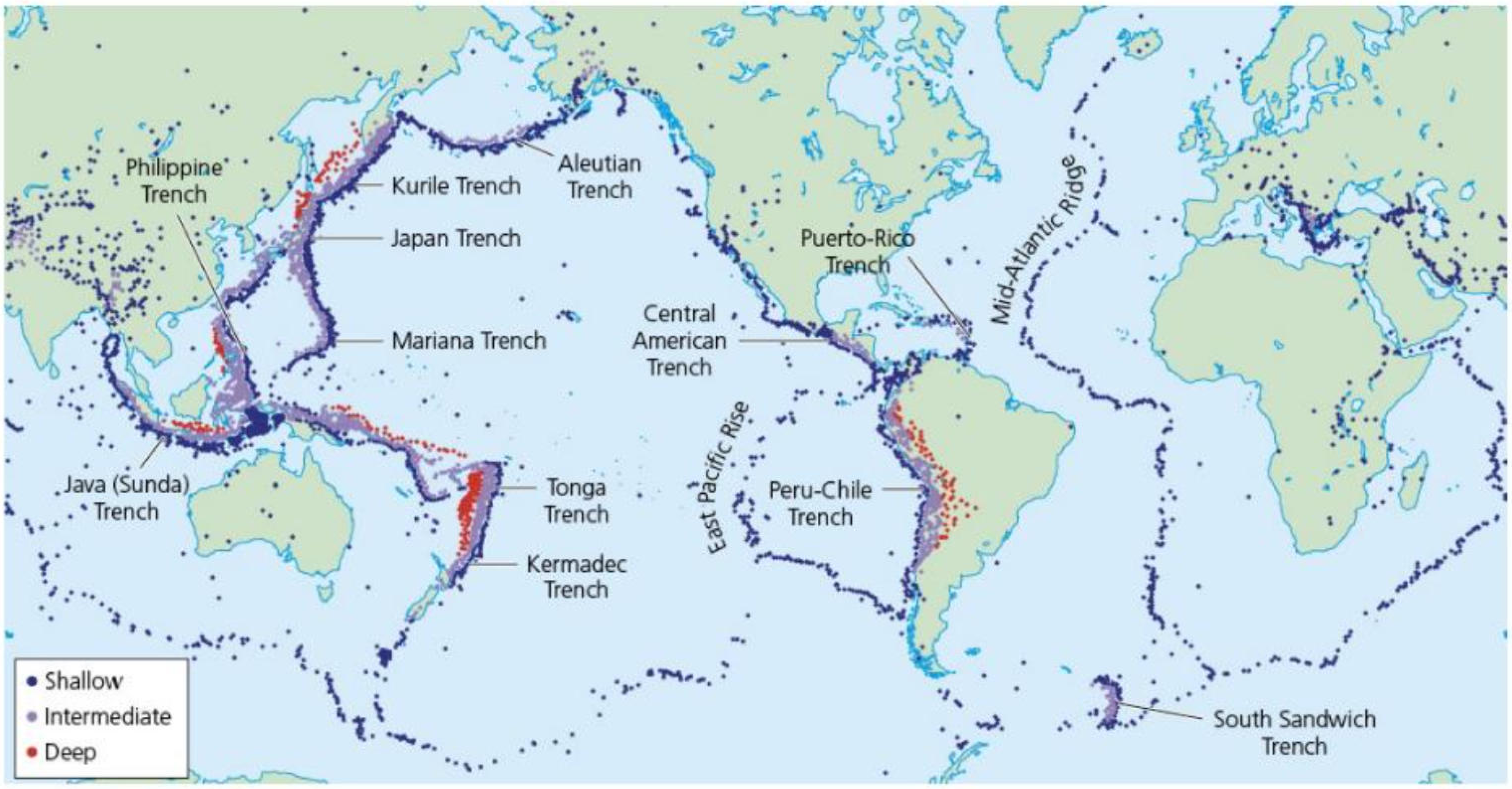
Seismic waves are the 'shock waves' generated by the release of energy following an earthquake. There are different types of seismic waves:

- **Body waves** spread out from the focus of the earthquake.
- **Surface waves** spread out from the epicentre. Surface waves are slower than body waves.

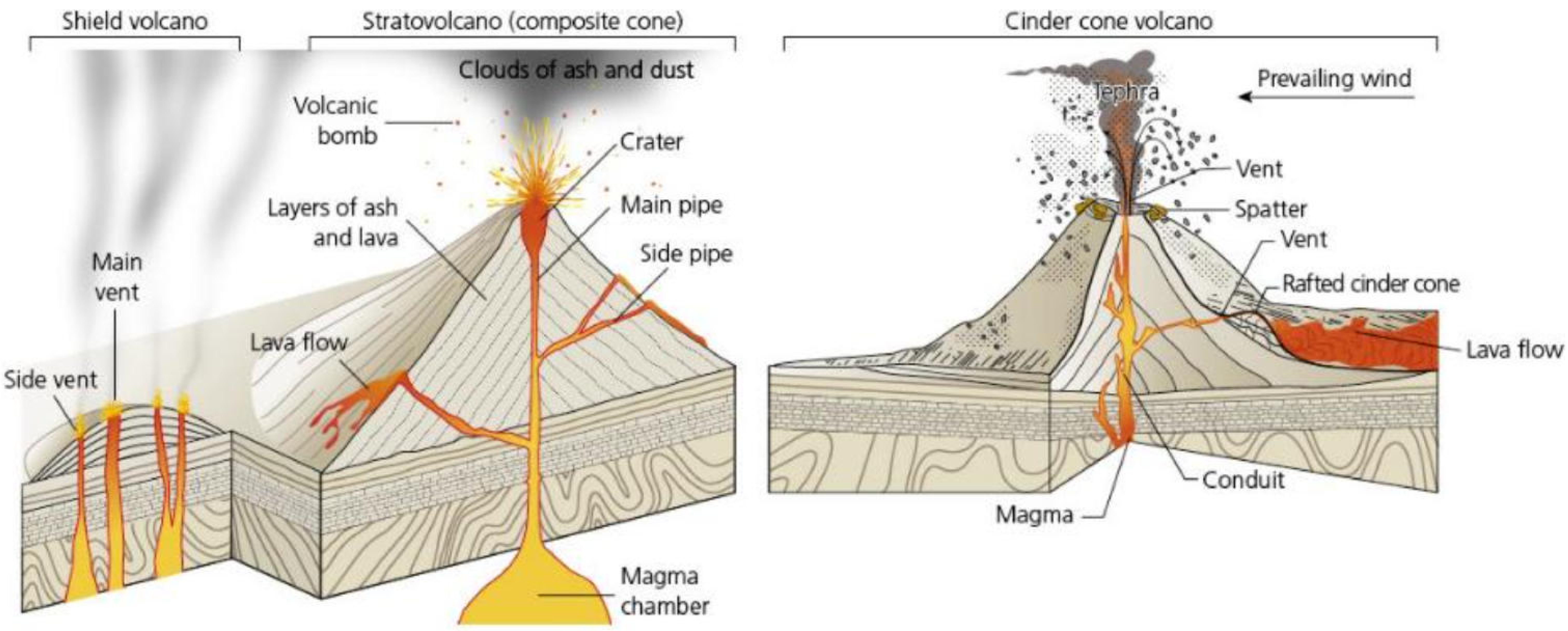
Body waves include **primary waves** (P-waves) and **secondary waves** (S-waves). Primary waves are compressional waves and are very fast (4–7 km/second). They can pass though liquids. In contrast, secondary waves are slower (2–5 km/second) and cannot pass through liquids.

Surface waves are produced when the energy from an earthquake reaches the Earth's surface and causes a rolling or swaying motion. They are the slowest type of wave, but they cause the most damage.

Scientists use these waves to determine the exact location of the epicentre of the earthquake.



▲ **Figure 4.10** The distribution of shallow-, intermediate- and deep-focus earthquakes, 1900–2017



▲ **Figure 4.11** Shield, stratovolcano (composite cone) and cinder cone volcanoes

Activities

1 Describe the distribution of shallow- and deep-focus earthquakes.
2 Distinguish between body waves and surface waves.

The main types of volcanoes

The shape of a volcano depends on the type of lava it contains.

» **Shield volcanoes** are gently sloping and produced by very hot, runny basaltic lava, for example Mauna Loa in Hawaii. They can cover a very wide area. They are made up of many layers of lava from repeated eruptions.
» **Cinder cone volcanoes** are steep-sided, conical hills produced by thicker, less fluid material. They are made up of loose, pyroclastic fragments built up around a volcanic vent. Most cinder cones have a crater at the summit. Parícutin in Mexico was formed by ash and cinders building up into a symmetrical cone as a result of eruptions in the 1940s.
» **Stratovolcanoes** (also known as **composite cone volcanoes**) are built up of many layers of hardened lava and tephra (volcanic ash, cinders, pumice and bombs), for example Mount Nyiragongo in Democratic Republic of the Congo and Mount Etna in Sicily. A stratovolcano is one that has had more than one eruption, and later eruptions may change the shape of the volcano. For example, Vesuvius was altered by the eruption in AD 79, forming a new version of itself, although some parts of the previous volcano survived the eruption. The shape of the volcano also depends on the amount of change there has been since the last volcanic eruption.

Cone volcanoes are generally associated with convergent/destructive plate boundaries, whereas shield volcanoes are characteristic of divergent/constructive boundaries and hotspots (areas of weakness in the middle of a plate).

Activities

1 Distinguish between shield and cone volcanoes.
2 Identify the types of plate boundaries associated with shield and cone volcanoes.

The classification of volcanoes

Active volcanoes are those that have erupted in recent times, such as Grindavik in Iceland, Kîlauea in Hawaii and Whakaari/White Island in New Zealand, which all erupted in 2023–24 and could erupt again.

Dormant volcanoes are volcanoes that have not erupted for many centuries but may erupt again, such as Mount Rainier in the USA.

Extinct volcanoes are not expected to erupt again. Kilimanjaro in Tanzania is an excellent example of an extinct volcano. The Le Puys region of France is an area of extinct volcanoes, which continue to influence settlements and tourism.

The main features of volcanoes

A **crater** is the depression at the top of a volcano following a volcanic eruption. It may contain a lake. A **vent** is the channel that allows magma within the volcano to reach the surface in a volcanic eruption. Magma is the molten material from the Earth's interior. The **magma chamber** refers to the reservoir of magma located deep inside the volcano. A **secondary cone** is a small volcano found on the side of the main volcano. For example, Mount Etna has several secondary cones (Figure 4.12).

▲ **Figure 4.12** Secondary cones on Mount Etna

The hazards associated with volcanoes

Lava flows are streams of molten rock that come from an erupting vent. They can be erupted from an explosive or nonexplosive event. The speed at which lava flows move across the ground depends on many factors, such as the type of lava erupted, the steepness of the ground, whether the lava flows as a broad feature or in a confined channel, and the rate of lava production at the vent. In 2023–24, lava flows in Grindavik, Iceland (Figure 4.13) caused much disruption and led to the death of one person.

▲ **Figure 4.13** Lava flows, Grindavik, Iceland, 2023–24

Tephra refers to small fragments of rock ejected into the air by an erupting volcano. Most tephra falls back on to the slopes of the volcano, so enlarging it. But billions of smaller and lighter pieces less than 2 mm in diameter (ash) can be carried hundreds of kilometres by winds. **Ash falls** are potentially very hazardous and can disrupt human activities across thousands of square kilometres.

Ash fallout to the ground can cause significant disruption and damage to buildings, transportation, water and wastewater, power supply, communications equipment, agriculture and primary production. This can lead to potentially substantial societal impacts and costs, even at thicknesses of only a few millimetres. Additionally, when ingested, the fine-grained ash can cause health impacts to humans and animals.

Volcanic ash clouds are a major hazard to aviation. The 2010 eruption of Eyjafjallajökull in Iceland had a major impact on air travel. Over 300 airports in Europe were closed between 15 and 21 April 2010. More than 100,000 flights were cancelled, affecting around 7 million passengers and causing about US$1.7 billion in lost revenue for the aviation industry.

A lahar is a fast-flowing mudflow made up of volcanic ash and water. The water may come from surface water (for example, from the lake in the volcano’s crater), rain or melting snow. Lahars flow down valleys and material is deposited over low-lying areas. Following the eruption of Mount Pinatubo in 1991, a tropical cyclone brought heavy rain and caused a number of lahars to form.

▲ **Figure 4.14** Pyroclastic flow on the Soufrière Hills volcano, Montserrat – a cone-shaped volcano formed by the subduction of the North American and South American Plates beneath the Caribbean Plate

Pyroclastic flows (Figure 4.14) are thick clouds of volcanic ash, pumice (volcanic rock) and gas that flow at high speed from a volcano into the valleys below. The temperature inside the pyroclastic flow can reach 700°C and flows can move at speeds of over 500 km/h. When they cool down, they form a layer of ash across the landscape that solidifies into pumice.

A **volcanic block** is a fragment of rock, over 64 mm in diameter, that is ejected in a solid condition from a volcano during an explosive eruption. Blocks are formed from material that was created during previous eruptions. In contrast, a volcanic bomb is a mass of partially molten rock (tephra) formed when a volcano ejects viscous fragments of lava during an eruption.

Volcanic gases may escape through vents in the ground. The most common volcanic gases include carbon dioxide, sulphur dioxide, hydrogen sulphide and carbon monoxide. Some of these gases are irritating or poisonous, or cause breathing problems. The release of sulphur dioxide may cause acid rain to form. The escape of carbon dioxide from Lake Nyos in Cameroon in 1986, possibly due to a small volcanic eruption, led to the death of over 1700 people and about 3500 livestock. This happened quickly so there was no warning and it was an infrequent event, but it was a high-magnitude event with a major impact. Fortunately, the gases did not spread to other areas.

Interesting notes

- ★ The greatest volcanic eruption in recorded history was Tambora in Indonesia in 1815. Some 50–80 km^3 of material was blasted into the atmosphere.
- ★ In 1883 the explosion of Krakatoa was heard from as far as 4776 km away.
- ★ The world's largest active volcano is Mauna Loa in Hawaii, which is 120 km long and over 100 km wide.

Activities

1. Distinguish between a volcano's crater and its chamber.
2. Explain three hazards related to volcanic activity.